

Condolence Letter & Final Hospital Summary

To: Dr. Samuel Cohen (Primary Care)

From: Hematology/Oncology Service

Date of letter: 20.11.2023

Patient: Cynthia Nelson, MRN AML044

Primary diagnosis: Acute Myeloid Leukemia, ELN 2022 Adverse risk, inv(3) (q21.3q26.2) with MECOM (EVI1) activation

Initial therapy: CPX-351 (daunorubicin/cytarabine liposomal) started 09.09.2022

Notification of Death

With great sadness we inform you that **Ms. Cynthia Nelson (*14.03.1970)** died on **09.11.2023** on our hematology service. Immediate causes were **diffuse alveolar hemorrhage and refractory hypoxemic respiratory failure** in the setting of **relapsed AML with profound thrombocytopenia** and **probable invasive pulmonary aspergillosis**. Code status was **DNR/DNI** following goals-of-care discussions with her family.

Diagnostic Background

- **Presentation (01.09.2022):** Fatigue, bruising, and weight loss (6 kg). Labs: **WBC $12.8 \times 10^9/L$** (blasts 24%), **Hgb 8.4 g/dL**, **Plt $41 \times 10^9/L$** , **LDH 610 U/L**.
 - **Bone marrow:** ~70% myeloblasts; flow CD34⁺/CD117⁺/HLA-DR variable; aberrant CD7.
 - **Cytogenetics:** inv(3)(q21.3q26.2); FISH positive for MECOM rearrangement. FLT3/NPM1 negative.
 - **Baseline echo:** LVEF 59%; CT chest without acute disease.
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Treatment Course

1. **1st-line CPX-351** (09.09.2022 induction; consolidation in Nov 2022). Day-35 marrow in October showed **partial response** with persistent disease (~8–10% blasts). She required prolonged transfusion support but had only one febrile episode (culture-negative).
2. **Progression:** Early **June 2023** bone marrow revealed **re-expansion to ~32% blasts** with worsening cytopenias and rising LDH. Circulating blasts reappeared.
3. **Salvage attempts (June–Sep 2023):**
 - **Azacitidine + venetoclax** initiated with inpatient ramp-up; **posaconazole** for mold prophylaxis (venetoclax dose reduced accordingly). After two cycles, marrow remained **>20% blasts**.
 - Screening for **clinical trial** of a **menin inhibitor** was completed; she was **ineligible** due to the absence of KMT2A/NRAS/KRAS targets and persistent fungal markers.

- Brief trial of **FLAG-IDA (attenuated)** in September was stopped early for worsening respiratory status and **platelets $<5 \times 10^9/L$** refractory to transfusion.
4. **Complications (Oct–Nov 2023):**
- **Probable invasive aspergillosis:** CT chest with halo sign lesions; serum **galactomannan positive** (index 1.2). Treated with **voriconazole** (levels therapeutic).
 - **Diffuse alveolar hemorrhage (DAH):** abrupt hemoptysis, hypoxemia, and diffuse ground-glass changes on high-resolution CT; bronchoscopy deferred due to instability.
 - Persistent **DIC-like picture:** fibrinogen fluctuating 120–280 mg/dL, D-dimer markedly elevated, INR up to 1.8; required cryoprecipitate and platelets.
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Final Admission (02.11.2023 → 09.11.2023)

Reason: Hemoptysis, dyspnea, and fever. On arrival: T 38.4°C, BP 98/60, HR 112, RR 28, SpO₂ 89% on room air.

Key labs: WBC $0.6 \times 10^9/L$ (blasts 6%), **ANC 0.03**, Hgb 6.7 g/dL, **Plt $4 \times 10^9/L$** ; Cr 1.0 mg/dL; AST/ALT 42/38; fibrinogen 140 mg/dL; INR 1.7; LDH 980 U/L.

Imaging: CT chest with bilateral diffuse ground-glass and nodular opacities; small pleural effusions.

Microbiology: Blood cultures negative; sputum grew **Aspergillus fumigatus** on fungal culture (later).

Management: High-flow nasal oxygen; **voriconazole** IV; broad-spectrum antibiotics (meropenem); **tranexamic acid nebulized**; platelet transfusions every 6–8 h with poor increments (allo-immunization suspected); cryoprecipitate for hypofibrinogenemia; cautious diuresis; aminocaproic acid considered but not pursued due to thrombotic risk.

Despite maximal supportive care, she developed escalating hypoxemia and tachypnea. After meeting with Ms. Nelson and her two adult children, we transitioned to comfort-focused care. She died peacefully at **04:28 on 09.11.2023** with family at bedside.

Comorbidities

- **Rheumatoid arthritis** (seropositive) previously on methotrexate (stopped at AML diagnosis); managed with intermittent low-dose prednisone during flares.
 - **Hypertension** (amlodipine 5 mg daily).
 - **Depression/anxiety** (sertraline 50 mg daily).
 - Former smoker; BMI 27.
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Medications Near End of Life

Voriconazole IV, meropenem, acyclovir, pantoprazole, PRN hydromorphone and lorazepam, ondansetron, stool regimen, frequent platelet and cryoprecipitate transfusions. Disease-directed chemotherapy had been stopped.

Aftercare

Next of kin **Daniel Nelson** (spouse) received bereavement resources. Autopsy declined. Death certificate completed and submitted; copies sent to your office.

Please accept our condolences. Contact us for any additional documentation.

Sincerely,
[Name], MD — Consultant Hematologist